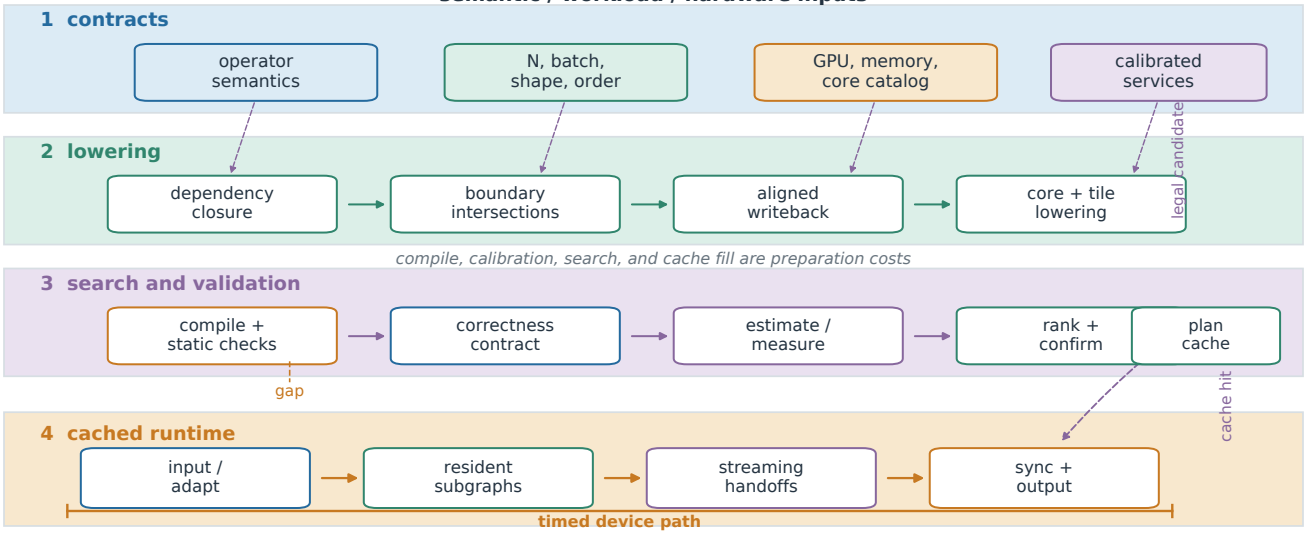

semantic / workload / hardware inputs
1 contracts
operator semantics
N, batch, shape, order
GPU, memory, core catalog
calibrated services
2 lowering
dependency closure
boundary intersections
aligned writeback
core + tile lowering
legal candidate
compile, calibration, search, and cache fill are preparation costs
3 search and validation
compile + static checks
correctness contract
estimate / measure
rank + confirm
plan cache
gap
cache hit
4 cached runtime
input / adapt
resident subgraphs
streaming handoffs
sync + output
timed device path
warm timing begins at device input and ends after the selected plan completes